

Problem Set for 27 March 2025

Exercise 0 Explain (without proving) the *implicit expressions*.

1. $\ker g$ as a subspace of $\ker fg$.

Solution. The linear map

$$\ker g \rightarrow \ker fg, \quad [x \mapsto x].$$

It is well-defined and injective.

2. $\text{im}(fg)$ as a quotient space of $\text{im}(g)$.

3. $\text{Hom}(V, W) \times \text{Hom}(U, V) \rightarrow \text{Hom}(U, W)$ the bi-linear map of composition

Solution. The bi-linear map

$$\text{Hom}(V, W) \times \text{Hom}(U, V) \rightarrow \text{Hom}(U, W), \quad (g, f) \mapsto g \circ f.$$

4. $\text{Hom}(U, V) \rightarrow \text{Hom}(U, W)$ induced by $f : V \rightarrow W$.

向后复合 f 的过程是线性映射:

$$\text{Hom}(U, V) \rightarrow \text{Hom}(U, W), \quad [U \xrightarrow{\varphi} V] \mapsto [U \xrightarrow{\varphi} V \xrightarrow{f} W].$$

5. $\text{Hom}(W, U) \rightarrow \text{Hom}(V, U)$ induced by $f : V \rightarrow W$.

向前复合 f 的过程是线性映射

$$\text{Hom}(W, U) \rightarrow \text{Hom}(V, U), \quad [W \xrightarrow{\varphi} U] \mapsto [V \xrightarrow{f} W \xrightarrow{\varphi} U].$$

6. $\text{Hom}_{\text{Sets}}(S, V) = V^n$ for finite set $|S| = n$.

考虑同构

$$\text{Hom}_{\text{Sets}}(S, V) \rightarrow V^n, \quad \begin{bmatrix} S & \xrightarrow{f} & V \\ s_i & \mapsto & f(s_i) \end{bmatrix} \mapsto (f(s_i))_{i=1}^n.$$

7. $\mathbb{F}[x]$ as a subspace of $\text{Hom}_{\mathbb{F}}(\mathbb{F}[[x]], \mathbb{F})$.

存在单射

$$\mathbb{F}[x] \mapsto \text{Hom}_{\mathbb{F}}(\mathbb{F}[[x]], \mathbb{F}), \quad \sum_{i=0}^n c_i x^i \mapsto \left[\begin{array}{ccc} \mathbb{F}[[x]] & \rightarrow & \mathbb{F} \\ \sum_{i=0}^{\infty} a_i y^i & \mapsto & \sum_{i=0}^n a_i c_i \end{array} \right].$$

8. The $\mathbb{C}$ -linear space $\text{Hom}_{\mathbb{R}}(\mathbb{C}, V)$, provided an $\mathbb{R}$ -linear space V .

对 $c \in \mathbb{C}$ 与 $f, g \in \text{Hom}_{\mathbb{R}}(\mathbb{C}, V)$ 构造

$$(cf + g) : \mathbb{C} \rightarrow V, \quad z \mapsto f(z \cdot c) + g(z).$$

9. (**Optional**) The set of all subsets of X forms a $\mathbb{F}_2$ -linear space, where $\mathbb{F}_2$ is a field with 2 elements.

构造双射

$$\text{Subset}(X) \rightarrow \text{Hom}_{\text{Sets}}(X, \mathbb{F}_2), \quad U \mapsto \left[\begin{array}{ccc} X & \rightarrow & \mathbb{F}_2 \\ U & \rightarrow & \{1\} \\ U^c & \rightarrow & \{0\} \end{array} \right].$$

此时 $\mathbf{0} = \emptyset$. 特别地, $\mathbb{F}_2$ 存在乘法结构, 从而 $\text{Hom}_{\text{Sets}}(X, \mathbb{F}_2)$ 也有乘法结构. 此处乘法即 $\cap$, 乘法单位元是 X .

这一双射也可以是

$$\text{Subset}(X) \rightarrow \text{Hom}_{\text{Sets}}(X, \mathbb{F}_2), \quad U \mapsto \left[\begin{array}{ccc} X & \rightarrow & \mathbb{F}_2 \\ U & \rightarrow & \{1\} \\ U^c & \rightarrow & \{0\} \end{array} \right].$$

此时 $\mathbf{0} = X$. 特别地, $\mathbb{F}_2$ 存在乘法结构, 从而 $\text{Hom}_{\text{Sets}}(X, \mathbb{F}_2)$ 也有乘法结构. 此处乘法即 $\cup$, 乘法单位元是 $\emptyset$.

这一构造 (前者常见) 在实分析中称作代数.

Example Construct a *satisfying* linear isomorphism

$$\text{Hom}_{\mathbb{F}}(U \times V, W) \simeq \text{Hom}_{\mathbb{F}}(U, W) \times \text{Hom}_{\mathbb{F}}(V, W).$$

Proof.

Step 1 Write down explicitly the linear map

$$\begin{aligned} \Phi : \quad \text{Hom}_{\mathbb{F}}(U, W) \times \text{Hom}_{\mathbb{F}}(V, W) &\rightarrow \text{Hom}_{\mathbb{F}}(U \times V, W) \\ (f, g) &\mapsto [(u, v) \mapsto f(u) + g(v)]. \end{aligned}$$

For a very detailed expression,

$$\begin{aligned} \Phi : \quad \text{Hom}_{\mathbb{F}}(U, W) \times \text{Hom}_{\mathbb{F}}(V, W) &\rightarrow \text{Hom}_{\mathbb{F}}(U \times V, W) \\ \left(\begin{bmatrix} U & \xrightarrow{f} & W \\ u & \mapsto & f(u) \end{bmatrix}, \begin{bmatrix} V & \xrightarrow{g} & W \\ v & \mapsto & g(v) \end{bmatrix} \right) &\mapsto \begin{bmatrix} U \times V & \xrightarrow{\Phi(f,g)} & W \\ (u, v) & \mapsto & f(u) + g(v) \end{bmatrix}. \end{aligned}$$

Step 2 Check that Φ is a well-defined linear map, i.e.

$$\Phi((f, g) + \lambda(f', g')) = \Phi(f, g) + \lambda\Phi(f', g'),$$

for both sides map (u, v) to $f(u) + f'(u) + \lambda g(v) + \lambda g'(v)$.

Step 3 Check that Φ is an injection. If $\Phi(f, g) = 0$, then so are the compositions

$$U \hookrightarrow U \times V \xrightarrow{\Phi(f,g)} W, \quad u \mapsto (u, 0) \mapsto f(u),$$

and

$$V \hookrightarrow U \times V \xrightarrow{\Phi(f,g)} W, \quad v \mapsto (0, v) \mapsto g(v).$$

It yields that $f = 0$ and $g = 0$.

Step 4 Check that Φ is a surjection. For any $\alpha : U \times V \rightarrow W$, consider

$$f_{\alpha} : U \rightarrow W, \quad u \mapsto \alpha(u, 0),$$

and

$$g_{\alpha} : V \rightarrow W, \quad v \mapsto \alpha(0, v).$$

Now $\Phi(f_{\alpha}, g_{\alpha}) = \alpha$, yielding that (f_{α}, g_{α}) is a preimage.

Step 5 Since Φ is well-defined, injective, and surjective, Φ is a linear isomorphism.

Your proof must include Step1-5.

Exercise 1 (optional if you have done this before) Show that $U \simeq \text{Hom}_{\mathbb{F}}(\mathbb{F}, U)$ for any $\mathbb{F}$ -linear space U .

Step 1 Write down explicitly the linear map $u \mapsto [1_{\mathbb{F}} \mapsto u]$, to be explicit,

$$\Phi : U \rightarrow \text{Hom}_{\mathbb{F}}(\mathbb{F}, U), \quad u \mapsto \begin{bmatrix} \mathbb{F} & \rightarrow & U \\ \lambda & \mapsto & \lambda u \end{bmatrix}.$$

Step 2 Check that Φ is a well-defined linear map, i.e., 对任意 $a \in \mathbb{F}$ 与 $u, v \in U$,

$$\Phi(u + av)(\lambda) = \lambda(u + av) = \lambda u + a(\lambda v) = (\Phi(u) + a\Phi(v))(\lambda).$$

Step 3 Show that Φ is injective, your solution...

若 $\Phi(u) = 0$, 也就是 $\{\lambda u\}_{\lambda \in \mathbb{F}} = \{0\}$, 当且仅当 $u = 0$.

Step 4 Show that Φ is surjective, your solution...

任意 $f \in \text{Hom}_{\mathbb{F}}(\mathbb{F}, U)$ 都有原像 $f(1)$, 因为

$$\Phi(f(1)) : \mathbb{F} \rightarrow U, \quad \lambda \mapsto f(1)\lambda = f(\lambda).$$

Step 5 Conclude that Φ is an linear isomorphism.

Exercise 2 (optional if you have done this before) Show that $U \simeq \text{Hom}_{\mathbb{F}}(\text{Hom}_{\mathbb{F}}(U, \mathbb{F}), \mathbb{F})$ if $\dim U < \infty$.

Step 1 Write down explicitly the linear map $u \mapsto [f \mapsto f(u)]$, to be explicit,

$$\Phi : U \rightarrow \text{Hom}_{\mathbb{F}}(\text{Hom}_{\mathbb{F}}(U, \mathbb{F}), \mathbb{F}), \quad u \mapsto \begin{bmatrix} \text{Hom}_{\mathbb{F}}(U, \mathbb{F}) & \rightarrow & \mathbb{F} \\ f & \mapsto & f(u) \end{bmatrix}.$$

Step 2 Check that Φ is a well-defined linear map, i.e., 即证明 u 处的赋值是线性函数:

$$\Phi(u)(\lambda f + g) = (\lambda f + g)(u) = \lambda(f(u)) + g(u) = \lambda(\Phi(u)(f)) + (\Phi(u))(g).$$

Step 3 Show that Φ is injective, your solution...

若 $\Phi(u) = 0$, 则所有 $\text{Hom}_{\mathbb{F}}(U, \mathbb{F})$ 中的函数在 u 处取值为 0. 由于 U 是有限维空间, 任取选定直和补 $U = \text{span}(u) \oplus U'$ 并构造线性映射

$$l : \text{span}(u) \oplus U' \rightarrow \mathbb{F}, \quad (\lambda u, v) \mapsto \lambda l(u).$$

依照 $\Phi(u) = 0$, 得 $l(u) = 0$. 此时 $u = 0$.

Step 4 Show that $\dim U = \dim \text{Hom}_{\mathbb{F}}(U, \text{Hom}_{\mathbb{F}}(U, \mathbb{F}))$ by induction.

若 $\dim U = n$, 则 $\text{Hom}_{\mathbb{F}}(U, \mathbb{F})$ 对应矩阵空间 $\mathbb{F}^{1 \times n}$, 维数亦为 n . 归纳地有

$$\dim U = \dim \text{Hom}_{\mathbb{F}}(U, \text{Hom}_{\mathbb{F}}(U, \mathbb{F})) = n.$$

Step 5 Conclude that Φ is an linear isomorphism by dimensional analysis.

Exercise 3 Let V be a linear space and $S \subset V$ is linearly independent (S is not necessary finite). Show that

$$\text{Hom}_{\mathbb{F}}(\text{Span}(S), \mathbb{F}) \simeq \text{Hom}_{\text{Sets}}(S, \mathbb{F}).$$

Step 1 Write down explicitly the linear map $f \mapsto (f(s))_{s \in S}$, to be explicit,

$$\Phi : \text{Hom}_{\text{Sets}}(S, \mathbb{F}) \rightarrow \text{Hom}_{\mathbb{F}}(\text{Span}(S), \mathbb{F}), \quad \left[\begin{array}{ccc} S & \xrightarrow{f} & \mathbb{F} \\ s_i & \mapsto & f(s_i) \end{array} \right] \mapsto \left[\begin{array}{ccc} \text{span}(S) & \rightarrow & \mathbb{F} \\ \sum_i^{\text{有限}} \lambda_i s_i & \mapsto & \sum_i^{\text{有限}} \lambda_i f(s_i) \end{array} \right]$$

Step 2 Check that Φ is a well-defined linear map, i.e., 以上给出了 $\Phi(f)$ 在 $\text{Span}(S)$ 的一组基 S 上的定义, 因此决定了唯一的线性映射.

Step 3 Show that Φ is injective, your solution...

若 $\Phi(f) = 0$, 则必有 $f(s_i) \equiv 0$. 此时 $f = 0$.

Step 4 Show that Φ is surjective, your solution...

对任意 $g : \text{Span}(S) \rightarrow \mathbb{F}$, g 由全体 $\{g(s_i)\}_{s_i \in S}$ 决定, 从而 $\Phi(g|_S) = g$.

Step 5 Conclude that Φ is an linear isomorphism.

In particular, $\mathbb{F}[[x]] \simeq \text{Hom}_{\mathbb{F}}(\mathbb{F}[x], \mathbb{F})$.

Exercise 4 Recall that $\mathbb{C}$ -linear spaces are automatically $\mathbb{R}$ -linear, and so are $\mathbb{C}$ -linear maps. Show that

$$\mathrm{Hom}_{\mathbb{R}}(U, V) \simeq (\mathrm{Hom}_{\mathbb{C}}(U, V))^2 \quad (\text{as } \mathbb{R}\text{-linear spaces}).$$

Here $W^2 := W \times W$.

- We assume that both U and V are of finite dimension, so that $\overline{U}$ is defined (i.e., $U = X \oplus i \cdot X$ for some X).

Step 0 Show that $\mathrm{Hom}_{\mathbb{C}}(U, V)$ is indeed an $\mathbb{R}$ -linear space.

$\mathrm{Hom}_{\mathbb{C}}(U, V)$ 是 $\mathbb{C}$ -线性空间, 从而是 $\mathbb{R}$ -线性空间.

Step 1 Write down explicitly the linear map $f \mapsto (f_1, f_2)$, to be explicit,

$$\Phi : \mathrm{Hom}_{\mathbb{R}}(U, V) \rightarrow (\mathrm{Hom}_{\mathbb{C}}(U, V))^2, \quad f \mapsto \left(\begin{bmatrix} U & \xrightarrow{f_1} & V \\ u & \mapsto & f_1(u) \end{bmatrix}, \begin{bmatrix} U & \xrightarrow{f_2} & V \\ u & \mapsto & f_2(u) \end{bmatrix} \right).$$

先明确 f_1 的构造. 对 $\mathbb{C}$ -线性空间之间的 $\mathbb{R}$ -线性函数 $f : U \rightarrow V$, 考虑

$$f_1 : U \rightarrow V, \quad u \mapsto f(u) - if(iu).$$

这一新函数是 $\mathbb{C}$ -线性的, 因为

$$f_1(iu) = f(iu) - if(i(iu)) = i(f(u) - if(iu)) = i(f_1(u)).$$

同理, $f_2(u) := \overline{f(u)} - i \cdot \overline{f(iu)}$ 也是 $\mathbb{C}$ -线性的. 此处仅是将 f 换作了 $\overline{f}$.

Step 2 Check that Φ is a well-defined linear map, i.e.,

依照积 Φ 线性当且仅当 Φ 在各分量线性. 换言之, $f \mapsto f_1$ 与 $f \mapsto f_2$ 均是 $\mathbb{R}$ -线性的. 直接验证得

$$(f + \lambda g)_1(u) = (f + \lambda g)(u) - i(f + \lambda g)f(iu) = f_1(u) + \lambda g_1(u),$$

以及 (注意 $\overline{\overline{\lambda}} = \lambda$)

$$(f + \lambda g)_2(u) = \overline{(f + \lambda g)(u)} - i\overline{(f + \lambda g)f(iu)} = f_2(u) + \lambda g_2(u).$$

从而 Φ 是 $\mathbb{R}$ -线性的.

Step 3 Show that Φ is injective, your solution...

若 $(f_1, f_2) = 0$, 则 $f = \frac{f_1 + \overline{f_2}}{2} = 0$.

Step 4 Show that Φ is surjective, your solution...

对任意 $f, g \in \mathrm{Hom}_{\mathbb{C}}(U, V)$, 构造 $\varphi = \frac{f + \overline{g}}{2}$.

$$\varphi_1(u) = \frac{f(u) + \overline{g(u)}}{2} - i \cdot \frac{f(iu) + \overline{g(iu)}}{2} = \frac{f(u) + \overline{g(u)}}{2} - i \cdot \frac{if(u) - i \cdot \overline{g(u)}}{2} = f(u).$$

类似地,

$$\varphi_2(u) = \frac{\overline{f(u) + \overline{g(u)}}}{2} - i \cdot \frac{\overline{f(iu) + \overline{g(iu)}}}{2} = \frac{\overline{f(u)} + g(u)}{2} - i \cdot \frac{-i\overline{f(u)} - i \cdot g(u)}{2} = g(u).$$

从而 Φ 满.

Step 5 Conclude that Φ is an linear isomorphism.

Optional Exercise Show that

$$\text{Hom}_{\mathbb{R}}(U, V) \simeq (\text{Hom}_{\mathbb{C}}(U, V))^2 \quad (\text{as } \mathbb{C}\text{-linear spaces}).$$

此处存在一个问题: $\text{Hom}_{\mathbb{R}}(U, V)$ 上的 $\mathbb{C}$ 线性结构有两种:

1. $(zf + g)(u) := f(zu) + g(u);$
2. $(zf + g)(u) := zf(u) + g(u).$

对一般 $\mathbb{C}$ -线性空间间的 $\mathbb{R}$ -线性函数 f , 不必有 $f(zu) = zf(u)$. 例如

$$\overline{(-)} : \mathbb{C} \rightarrow \mathbb{C}, \quad z \mapsto \bar{z}$$

是 $\mathbb{R}$ -线性的, 但不见得 $\overline{zu} = z\bar{u}$.

一个快捷的方式是将 $\mathbb{C}$ -线性空间写作两个 $\mathbb{R}$ -线性空间的直和:

$$U = X \oplus iX, \quad V = Y \oplus iY.$$

从而

$$\text{Hom}_{\mathbb{R}}(X \oplus iX, Y \oplus iY) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \text{Hom}_{\mathbb{R}}(X, Y) \right\}.$$

特别地, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ 是 $\mathbb{C}$ -线性映射当且仅当其形如 $\begin{pmatrix} a & -c \\ c & a \end{pmatrix}$. 以下分别给出 $\text{Hom}_{\mathbb{R}}(U, V)$ 上两种 $\mathbb{C}$ 线性结构对应至 $(\text{Hom}_{\mathbb{C}}(U, V))^2$ 的方式

1. $\text{Hom}_{\mathbb{R}}(X \oplus iX, Y \oplus iY) \simeq \text{Hom}_{\mathbb{R}}(X, Y \oplus iY)^2 \simeq \text{Hom}_{\mathbb{C}}(X \oplus iX, Y \oplus iY)^2;$
 - 前一同构: 显然
 - 后一同构: X 至 V 的 $\mathbb{R}$ -线性映射可以唯一地提升做 U 至 V 的 $\mathbb{C}$ -线性映射.
2. $\text{Hom}_{\mathbb{R}}(X \oplus iX, Y \oplus iY) \simeq \text{Hom}_{\mathbb{R}}(X \oplus iX, Y)^2 \simeq \text{Hom}_{\mathbb{C}}(X \oplus iX, Y \oplus iY)^2.$
 - 前一同构: 显然
 - 后一同构: 参考 $f \mapsto f_1$.